Def. Let V be a vector space.
Given a vector $\alpha \in V$ and linear operator $T: V \to V$, the subspace
$$Z(\alpha, T) \overset{\text{def}}{=} \{ g(T)\alpha \mid g(t) \in F[t] \} = \operatorname{span}\{ T^k \alpha \mid k \in \mathbb{N} \cup \{0\} \}.$$
is called the T-cyclic subspace generated by α

Def. Let V be a vector space. Given $\alpha \in V$ and a linear operator T on V, an ideal
$$M(\alpha, T) = \{ g(t) \in F[t] \mid g(T)\alpha = 0 \}$$
is called the T-annihilator of α, and the unique monic generator of also called ".

Thm. Let α be any non-zero vector in V and let p_α be the T-annihilator of α.
1) $\deg p_\alpha = \dim Z(\alpha, T)$
2) $Z(\alpha, T) = \operatorname{span}\{ T^k \alpha \mid 0 \le k \le \deg p_\alpha \}$
3) The minimal polynomial of $T|_{Z(\alpha,T)}$ is p_α.
Proof. Given a polynomial $g \in F[x]$, the Euclidean Algorithm gives
$$g = p_\alpha \cdot q + r \quad \text{for some } q, r \in F[x], \text{ with } r = 0 \text{ or } \deg r < \deg p_\alpha.$$
Then, $g(T)\alpha = p_\alpha(T) q(T)\alpha + r(T)\alpha = r(T)\alpha$. Since either $r=0$ or $\deg r < \deg p_\alpha$, the vector $r(T)\alpha$ is a linear combination of $\{\alpha, T\alpha, T^2\alpha, \cdots, T^{\deg p_\alpha - 1}\alpha\}$.
Since g was arbitrary, $Z(\alpha, T)$ is spanned by $\{ T^n \alpha \mid n = 0, 1, \cdots, \deg p_\alpha - 1 \}$
Moreover, these are linearly independent, because: if there exists non-trivial representation
$$c_0 \alpha + c_1 T\alpha + \cdots + c_{\deg p_\alpha - 1} T^{\deg p_\alpha - 1} \alpha = 0,$$
then it contradicts to the minimality of p_α. It proved 1) and 2).
To show the third assertion, let $U = T|_{Z(\alpha,T)} : Z(\alpha, T) \to Z(\alpha, T) : \alpha \mapsto T\alpha$.
It is well-defined, since the cyclic subspace is invariant under T.
Given a polynomial $g \in F[x]$,
$$p_\alpha(U) g(T)\alpha \underset{\substack{\uparrow \\ g(T)\alpha \in Z(\alpha, T)}}{=} p_\alpha(T) g(T)\alpha = g(T) p_\alpha(T)\alpha = g(T)0 = 0.$$
So, $p_\alpha(U)$ is identically zero. Moreover, $h(U)$ cannot be zero if $\deg h < \deg p_\alpha$, because $h(U)\alpha = h(T)\alpha = 0$, then it contradicts to the minimality.

「Companion Matrix.

Let W be a vector space of dimension $k \in \mathbb{N}$, and $T: W \to W$ be a linear operator which have a cyclic vector $\alpha \in W$.
By the Theorem above, $\{\alpha, T\alpha, \cdots, T^{k-1}\alpha\}$ forms a basis for W.
As above fact and the Cayley-Hamilton Theorem, let

$$p_\alpha(x) = x^k + c_{k-1}x^{k-1} + \cdots + c_1 x + c_0$$

be the minimal polynomial (and also characteristic) for T.
Define $\beta = \{T^{n-1}\alpha \mid n = 1, 2, \cdots, k\}$, denote $\alpha_n = T^{n-1}\alpha$. Then,

$$0 = 0 \cdot \alpha = p_\alpha(T)\alpha = T^k\alpha + c_{k-1}T^{k-1}\alpha + \cdots + c_1 T\alpha + c_0\alpha$$

$$\Rightarrow T^k\alpha = -c_0\alpha - c_1 T\alpha - \cdots - c_{k-1}T^{k-1}\alpha$$
$$= -c_0\alpha_1 - c_1\alpha_2 - \cdots - c_{k-1}\alpha_k$$

$$[T]_\beta = \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & -c_{k-1} \end{bmatrix} = \begin{bmatrix} 0 & \cdots \quad 0 & -c_0 \\ & & \vdots \\ & I_{k-1} & \vdots \\ & & -c_{k-1} \end{bmatrix} ..$$

Directly, we obtain that: on a fin. dim space T,

A linear operator T has a cyclic vector

if and only if

the minimal polynomial of T and characteristic polynomial of T are identical.

⌈Exercises in Hoffman, Sec 7.2

1. Every linear operator T on two-dimensional either;

T has a cyclic vector or T is a scalar multiple of the identity.

Proof. Suppose that T has no cyclic vector. That is,